

The Mechanical Vacuum: Frame-Dragging, Satiation Velocity, and the Variable Gravitational Constant

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Abstract

In standard General Relativity (GR), G is a coupling constant that determines how much spacetime curves for a given amount of energy-momentum. GR treats G as a static scalar, whereas this model treats it as a dynamic equilibrium state of the field itself. GR views G in Einstein's field equations¹ (typically written) as:

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4}T_{\mu\nu}$$

In this view, G is an “external” value. It is not a property of the field that reacts to the tensor $T_{\mu\nu}$; it is a fixed “conversion factor.” GR does not consider G to be in equilibrium with the tensor strength because G is assumed to be a fundamental property of the universe’s “hardware,” not a software-like variable that adjusts to local stress.

2. The Equilibrium Perspective (This Model) suggests that the “constant” G is actually the restorative response of the vacuum field. If we define the tensor strength as the “load” on the vacuum, then G must be in equilibrium with that load. Standard GR: G is a constant. If $T_{\mu\nu}$ (matter) increases, the curvature ($R_{\mu\nu}$) increases linearly because the “stiffness” (G) is fixed. In this model: G is a variable resultant. As $T_{\mu\nu}$ increases, the Vacuum Impedance increases, effectively “damping” or “stretching” the field’s ability to respond.

3. Historically, Einstein did consider the idea that the inertia of a body depends on the presence of other masses (Mach’s Principle), which would imply a variable gravitational influence. However, to keep the mathematics of General Relativity generally covariant and consistent with the Equivalence Principle, he settled on G as a universal constant. We expand upon GR by suggesting that G is in equilibrium with the tensor strength, the stiffness of spacetime is not uniform. It is “softer” where matter is dense (high impedance) and “stiffer” where matter is sparse (low impedance, high field tension).

This paper builds upon the original paper that described the mathematics of deriving the gravitational constant G as a function of vacuum restorative stiffness and matter-energy density²: $G = c^2/(8\pi\rho_m)$.

We describe that the observed missing mass in galactic rotation curves is a result of the vacuum recovering its ground-state tension in low-density environments. We provide a mathematical description for field-strain satiation and propose a testable prediction regarding gravitational lensing discrepancies.

1 The Mechanical Basis of Gravitation

In standard relativistic astrophysics, G is treated as a universal constant. However, applying a mechanical framework to the vacuum suggests that G represents the coupling efficiency between matter and the elastic fabric of space. We define c^2 as the restorative

stiffness of the vacuum medium. Matter-energy density ρ_m acts as the mechanical load, leading to the derivation:

$$G = \frac{c^2}{8\pi\rho_m} \quad (1)$$

2 Impedance, Strain, and Field-Strain Satiation

We define the vacuum as an elastic medium subject to impedance and strain.

2.1 Description of Terms

- ρ_m : Matter-Energy Density (The mechanical load).
- c^2 : Restorative Stiffness (The intrinsic modulus of the vacuum).
- Z_v : Vacuum Impedance (Opposition to force).
- ϵ : Field Strain (Deformation where $\epsilon \propto 1/G$).

2.2 Mathematical Correlation

In wave mechanics, impedance Z_v is proportional to ρ_m . As ρ_m increases, Z_v rises, leading to **Strain Satiation**. This occurs when the field is mechanically saturated and the coupling G approaches its minimum.

3 Prediction for Galactic Rotation Curves

The Dark Matter effect is reinterpreted as the recovery of the gravitational constant in low-impedance environments. Local measurements of G in high-density solar systems are suppressed. As ρ_m drops in the galactic outskirts, G increases toward its ground-state maximum, providing the necessary centripetal force to maintain stellar orbits.

4 Mechanical Validation and Scaling

The transition from local suppressions to universal maxima can be categorized into three pillars of evidence:

1. The Local Evidence (Mechanical Satiation)

In high-density environments like the Earth's surface, the vacuum is "loaded" by mass. This creates a state of mechanical satiation where G is at its minimum observed value (6.67×10^{-11}). The resulting velocity $v_\gamma \approx 0.00004$ m/s matches the order of magnitude of linear frame-dragging (Lense-Thirring effect) measured by Gravity Probe B³, identifying this velocity as the "viscosity" of the local vacuum frame.

2. The Universal Floor (Unruh Satiation)

The theoretical noise floor of the vacuum, as described by the Unruh Effect at $1g^4$, suggests a minimum thermal jitter. Our result of $40 \mu\text{m/s}$ identifies the propagation floor of the vacuum. This "clamping" effect prevents the gravitational coupling from diverging in the presence of planetary mass, maintaining the Newtonian limit.

3. The Galactic Solution (Dark Matter)

The variance between the local satiation velocity and c represents the dynamic range of the vacuum. As $\rho_m \rightarrow 0$ in galactic halos, the vacuum recovers its restorative stiffness, allowing G to return to its ground-state maximum. This increased coupling provides the "extra" centripetal pull required to flatten rotation curves without the need for additional dark mass.

5 Discussions

This mechanical model predicts that lensing in low-density voids is enhanced. This hypothesis aligns with theories proposed by Christodoulou and Kazanas (2019)⁵. Furthermore, studies of Void Lensing Anomalies (Sanchez et al., 2017)⁶ have noted discrepancies in lensing signals from certain cosmic voids, suggesting a possible variance in vacuum impedance.

A Technical Appendix

A.1 A1: Derivation of Velocity-Gravity Equivalence

The core of this framework lies in the link between the gravitational coupling constant (G) and the propagation velocity of light (v_γ).

Step 1: Foundational Mechanical Expression

We begin with G as a measure of field tension relative to density:

$$G = \frac{c^2}{8\pi\rho_m} \implies \rho_m = \frac{c^2}{8\pi G} \quad (2)$$

Step 2: Defining Vacuum Refractive Index

Applying the Lorentz-Lorenz principle to the vacuum medium, the refractive index n_v is determined by the mechanical load ρ_m :

$$n_v = \sqrt{1 + \rho_m} \quad (3)$$

Step 3: Deriving the Phase Velocity

Substituting the expression for ρ_m into the velocity equation $v_\gamma = c/n_v$:

$$v_\gamma = \frac{c}{\sqrt{1 + \frac{c^2}{8\pi G}}} = c\sqrt{\frac{8\pi G}{8\pi G + c^2}} \quad (4)$$

Step 4: Numerical Evaluation

To evaluate the expression, we use standard SI values ($c \approx 2.9979 \times 10^8$ m/s, $G \approx 6.6743 \times 10^{-11}$ m³ kg⁻¹ s⁻²):

$$8\pi G \approx 1.677 \times 10^{-9}$$

$$c^2 \approx 8.987 \times 10^{16}$$

Plugging these into the ratio:

$$\frac{8\pi G}{8\pi G + c^2} \approx 1.866 \times 10^{-26}$$

Taking the square root and multiplying by c :

$$v_\gamma \approx 0.00004 \text{ m/s}$$

A.2 Ground-State Restoration Table

Environment	Density	Effective G	Rotation Effect
Solar System	High	G-standard	Newtonian
Galactic Disk	Medium	G > G-standard	Partial Flatness
Galactic Halo	Low	G → G-max	Flat Rotation
Cosmic Void	→ 0	G-max	Max Lensing

Author Information

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